For instance, devices like Huawei Mate 20 Pro, Huawei Mate 30 Pro, Samsung Galaxy S10 and Samsung Galaxy Note 10+ offer this low-power solution and sizeable battery capacities to charge other Qi-compatible products including other smartphones, cameras, smartwatches, wireless headphones and so on. Different devices from various brands charge at different speeds.

Before activating this feature, protective covers and other accessories, particularly metal objects, need to be detached from both devices for proper working. Both devices need to be turned on. When the recipient device is paired back-to-back with the source device in a way that charging coils present inside them get aligned for proper connection, power starts getting transferred on its own. In some cases, multiple induction coils are present to ensure that the battery gets charged irrespective of the position of devices.

The screen confirms once the devices have connected and an LED indicator shows when the recipient phone begins charging, in most cases. Once required power is transferred, the devices can be separated. Only one recipient device can be charged at a time.

Source devices have a depletion limit on how much energy they can output from their batteries, such as disabling the option below twenty per cent.

Backs of source and recipient devices must be free from dust, dirt or moisture. Also, these should not be placed on a metal surface while charging. Charging efficiency depends on distance between coils, and condition of the devices or that of the surrounding environment. Charge transfer can also be done while the source phone is getting charged.

It is possible to mould the idea of using one device to charge another for different kinds of applications that require power. To understand the implications, take the example of US-based electric vehicle (EV) startup Rivian that plans to have a vehicle-to-vehicle (V2V) charging feature for its R1T and R1S models, in addition to auxiliary battery packs. However, it is yet to be seen how decentralised charging of this sort impacts the automobile sector.

—Ayushee Sharma

2 SOLAR POWER USE ON THE RISE FOR MOBILITY

Indian Railways is planning to become the world's first hundred per cent green operator within the next ten years

Several developments in the past decade are helping reduce dependence on non-renewable sources of energy in various sectors across the world. Great strides in renewable energy like solar and wind have been significant in reducing carbon footprints. When it comes to the automotive sector, electric vehicles (EVs) are already on the rise. Using solar energy can be a beneficial option for EVs especially for countries like India, which lie in the tropical region.

A solar vehicle can be fully or partially (hybrid or electric) powered by solar energy along with alternatives installed to deal with emergencies. Working is similar to any other application of solar energy for power generation. When sunlight falls on solar panels installed on the vehicle, solar energy gets converted to electricity and powers the car's motor directly or through batteries.

In addition to propulsion, solar energy can also be used for such functions as lighting, ventilation and air-conditioning in the vehicle through direct or indirect charging. This can be done without replacing the conventional engine. Many vehicles come with solar panels for optional mounting and monitoring systems for parameters like energy consumption and solar energy capture.

Solar panels for mounting on rooftops of buildings or other structures for power generation are already in high demand. According to a report by *MarketResearchNest.com*, the global distributed generation (DG) rooftop solar photovoltaic (PV) market is expected to expand at a CAGR of 7.3 per cent within the forecast period from 2019 to 2025. Such companies as Jinko Solar, SunPower, Solarworld and Kyocera Solar are among the major competitors in this market.

Growth in mobile applications can also be seen in a large number of train networks. In 2017, a fully solar-powered train was launched in New South Wales, Australia.

Indian Railways is planning to become the world's first hundred per cent green operator within the next ten years. Solar panels have already been installed on the rooftop of many coaches and tracksides, and static rooftop installations have further been planned to increase the solar generation capacity to 5GW by 2025. Also, Delhi Metro Rail Corp. (DMRC) has plans to build a hundred per cent solar-energy-dependent metro by 2021 through the support of Madhya Pradesh's Rewa Solar Project.

Besides being environmentally-friendly, there are no transmission losses or failures from natural disasters, as no traditional infrastructure is needed for battery-charging through solar. An EV may use solar as a secondary charging option to plug-in charging and increase its feasibility. Lightyear One, whose production is scheduled to start in 2021, is one such prototype EV that has solar panels and is aimed to provide up to 725km range.

While opportunities seem attractive, a number of limitations inhibit solar's commercial growth. Solar is yet in the nascent stage and people are skeptical about it. As large solar panels are required to generate power for the entire vehicle, weight and area of panels on rooftops become constraints in design.

Batteries used for storing energy so that it is available all day further increase the cost and weight of the